

Algebraic Geometry – Exercise Sheet 1

Week 1: Affine algebraic sets, projective space and localisation
Monday and Thursday afternoon sessions; Lectures 1–5

Difficulty. ★ routine; ★★ synthesis; ★★★ challenge or extension.

Use. The sheet is deliberately longer than the exercise sessions. Cerulean exercises form the suggested route; green exercises are alternatives.

Monday afternoon — material available: Lecture 1 only

Exercise 1. Three points in the affine plane

★ **CORE**

Let

$$S = \{(0, 0), (1, 0), (0, 1)\} \subset \mathbb{k}^2.$$

(i) Show that

$$\mathbf{I}(S) = (x, y) \cap (x - 1, y) \cap (x, y - 1).$$

(ii) Prove directly that

$$\mathbf{I}(S) = (xy, x(x - 1), y(y - 1)).$$

(iii) Show that evaluation at the three points induces an isomorphism

$$\mathbb{k}[x, y]/\mathbf{I}(S) \simeq \mathbb{k}^3.$$

Exercise 2. Two axes and their coordinate rings

★ **CORE**

In $\mathbb{k}^2$, set $X = \mathbf{V}(x)$ and $Y = \mathbf{V}(y)$.

(i) Show that

$$X \cup Y = \mathbf{V}(xy), \quad X \cap Y = \mathbf{V}(x, y).$$

(ii) Compute the coordinate rings of X , Y , $X \cup Y$, and $X \cap Y$.

(iii) Explain geometrically why $\mathbb{k}[x, y]/(xy)$ has zero divisors.

Exercise 3. The parabola and the cusp

★★ **CORE**

Consider

$$P = \mathbf{V}(y - x^2), \quad C = \mathbf{V}(y^2 - x^3) \subset \mathbb{k}^2.$$

(i) Show that $t \mapsto (t, t^2)$ defines an isomorphism $\mathbb{k} \rightarrow P$.

(ii) Show that $t \mapsto (t^2, t^3)$ defines a morphism $\nu : \mathbb{k} \rightarrow C$ which is bijective on the underlying sets.

(iii) Write the corresponding homomorphism of coordinate rings and prove that ν is not an isomorphism.

Exercise 4. Automorphisms of the affine and rational line

★★ **CHOICE**

(i) Show that morphisms of affine algebraic sets $\mathbb{k} \rightarrow \mathbb{k}$ are exactly polynomial maps $t \mapsto f(t)$. Describe the coordinate-ring homomorphism and check explicitly how composition reverses arrows.

(ii) Prove that such a morphism is an automorphism exactly when $f(t) = at + b$ with $a \in \mathbb{k}^\times$. Deduce that

$$\mathrm{Aut}_{\mathbb{k}\text{-alg}}(\mathbb{k}[t]) \simeq \mathbb{k} \rtimes \mathbb{k}^\times.$$

(iii) After projective space has been introduced, show that

$$t \mapsto \frac{at + b}{ct + d}, \quad ad - bc \neq 0,$$

defines a $\mathbb{k}$ -automorphism of $\mathbb{k}(t)$ and that scalar multiples of the matrix give the same automorphism. Check the order under composition. Using the standard fact $[\mathbb{k}(t) : \mathbb{k}(r)] = \deg(r)$ for a non-constant rational function r , prove that

$$\mathrm{Aut}_{\mathbb{k}\text{-alg}}(\mathbb{k}(t)) \simeq \mathrm{PGL}_2(\mathbb{k}).$$

Which subgroup preserves the subring $\mathbb{k}[t]$?

Exercise 5. Equations, zero sets and ideals of subsets★★ **CORE**Work over $\mathbb{R}$.

- (i) Show that, for every
- $m \geq 2$
- , the three distinct ideals

$$(x, y), \quad (x^2, xy, y^3), \quad (x, y)^m$$

define the same subset $\{(0, 0)\} \subset \mathbb{R}^2$. Determine the ideal of polynomials vanishing on this subset.

- (ii) Let
- $C = \{\pm 1\}^n \subset \mathbb{R}^n$
- be the set of vertices of the
- n
- dimensional cube. Prove that

$$\mathbf{I}(C) = (x_1^2 - 1, \dots, x_n^2 - 1).$$

- (iii) Regard
- $\mathbb{Z}$
- as a subset of
- $\mathbb{R}$
- . Compute
- $\mathbf{I}(\mathbb{Z}) \subset \mathbb{R}[t]$
- . Deduce that
- $\mathbb{Z}$
- is not an algebraic subset of the affine line over
- $\mathbb{R}$
- .

Exercise 6. Polynomial systems versus linear systems★ **CORE**Fix $n, d \geq 1$.

- (i) Recall why an invertible linear system of
- n
- equations in
- n
- unknowns has exactly one solution.
-
- (ii) Over
- $\mathbb{C}$
- , determine all solutions of

$$x_1^d = 1, \quad x_2^d = 1, \quad \dots, \quad x_n^d = 1.$$

Show that there are exactly d^n solutions.

- (iii) Write a system of
- n
- polynomial equations in
- n
- variables having no solution over
- $\mathbb{R}$
- .
-
- (iv) Explain precisely which expectations imported from linear algebra have failed in (ii) and (iii).

Thursday afternoon — material available: Lectures 1–4**Exercise 7. Radicals and reduced rings**★ **CORE**

Compute

$$\sqrt{(x^2, y^3)}, \quad \sqrt{(x^2 y^3)}$$

in $\mathbb{k}[x, y]$. Decide which of the rings

$$\mathbb{k}[x]/(x^2), \quad \mathbb{k}[x, y]/(xy), \quad \mathbb{k}[x, y]/(x^2, xy)$$

are reduced.

Exercise 8. Points at infinity★ **CORE**In the affine chart $x_0 \neq 0$ of $\mathbf{P}_{\mathbb{k}}^2$, write $x = x_1/x_0$ and $y = x_2/x_0$.

- (i) Homogenise the parabola
- $y = x^2$
- and determine its points at infinity.
-
- (ii) Homogenise the hyperbola
- $xy = 1$
- and determine its points at infinity.
-
- (iii) Explain geometrically why the answers are different.

Exercise 9. The line through two points of projective space★ **CHOICE**Let $p = [a_0 : \dots : a_n]$ and $q = [b_0 : \dots : b_n]$ be distinct points of $\mathbf{P}_{\mathbb{k}}^n$. Show that the line through them is defined by the 3×3 minors of

$$\begin{pmatrix} a_0 & \cdots & a_n \\ b_0 & \cdots & b_n \\ x_0 & \cdots & x_n \end{pmatrix}.$$

Write the equation explicitly for $n = 2$.**Exercise 10. Points in $\mathbf{P}^2$ from minors**★★ **CHOICE**Show that the 2×2 minors of

$$M = \begin{pmatrix} x & 0 \\ y & y \\ 0 & z \end{pmatrix}$$

define the three coordinate points in $\mathbf{P}^2$. For four lines $\ell_1, \dots, \ell_4$ with no three concurrent, show that the maximal minors of

$$\begin{pmatrix} \ell_1 & 0 & 0 \\ -\ell_2 & \ell_2 & 0 \\ 0 & -\ell_3 & \ell_3 \\ 0 & 0 & -\ell_4 \end{pmatrix}$$

define their six pairwise intersection points.

Exercise 11. Equations of the Segre and Veronese images

*** OPTIONAL

a) For

$$([s_0 : s_1], [t_0 : t_1]) \mapsto [s_0 t_0 : s_0 t_1 : s_1 t_0 : s_1 t_1],$$

compute the kernel of $z_{ij} \mapsto s_i t_j$ and identify the image in $\mathbf{P}^3$.

b) For the quadratic Veronese map $\mathbf{P}^2 \rightarrow \mathbf{P}^5$, arrange the target coordinates as a symmetric 3×3 matrix and show that the image is defined by its 2×2 minors.

Exercise 12. Localising the coordinate axes

★ CORE

Let

$$A = \mathbb{k}[x, y]/(xy).$$

- (i) Compute A_x and A_y .
- (ii) Compute $A_{(x)}$ and $A_{(y)}$.
- (iii) Explain what these localisations retain and what they discard geometrically.

Exercise 13. Localisation with a nilpotent

★ CHOICE

Let $A = \mathbb{k}[\varepsilon]/(\varepsilon^2)$. Compute

$$A_\varepsilon, \quad A_{(\varepsilon)}, \quad A_{1+\varepsilon}.$$

Explain why the three answers are so different.

Exercise 14. Two finite module structures

★★ CHOICE

Let

$$B = \mathbb{k}[x, y]/(y^2 - x).$$

Show that B is free of rank 2 over $\mathbb{k}[x]$ and free of rank 1 over $\mathbb{k}[y]$. Compare geometrically the corresponding morphisms to the affine line.

Exercise 15. Noether normalisation for the hyperbola

*** CHOICE

Let

$$B = \mathbb{k}[x, y]/(xy - 1), \quad u = x + y \in B.$$

Show that B is finite over $\mathbb{k}[u]$. Deduce that the corresponding morphism to the affine line $\mathbb{k}$ has fibres with at most two points.

After Friday's lecture or homework

Exercise 16. A presheaf which is not a sheaf

★ OPTIONAL

Let $X = U \amalg V$ be the disjoint union of two non-empty open subsets. Define a presheaf $\mathcal{F}$ by

$$\mathcal{F}(W) = \mathbb{Z} \quad \text{for } W \neq \emptyset, \quad \mathcal{F}(\emptyset) = 0,$$

with the evident restriction maps. Show that $\mathcal{F}$ is not a sheaf.

Exercise 17. Continuous and holomorphic functions

★★ CHOICE

Let $U \subset \mathbb{C}$ be connected and contain at least two points. Show that $\mathcal{O}(U)$ has no zero divisors, whereas $\mathcal{C}^0(U)$ has non-zero functions f, g with $fg = 0$.

Exercise 18. $\mathcal{O}(d)$ on the projective line

*** OPTIONAL

Identify $\mathbf{P}_{\mathbb{C}}^1$ with $\mathbb{C} \cup \{\infty\}$. Define $\mathcal{O}(d)$ as the sheaf of meromorphic functions which are holomorphic away from ∞ and have at most a pole of order d there when $d \geq 0$; for $d < 0$, require a zero of order at least $-d$ at ∞ . Compute its global sections.

Algebraic Geometry – Exercise Sheet 2

Week 2: Prime spectra, the structure sheaf and points of affine space

Monday and Thursday afternoon sessions; Lectures 6–10

Difficulty. ★ routine; ★★ synthesis; ★★★ challenge or extension.

Use. The sheet is deliberately longer than the exercise sessions. Cerulean exercises form the suggested route; green exercises are alternatives.

For a ring R , write $\gamma_R : \mathbb{Z} \rightarrow R$, $n \mapsto n1_R$, for the unique unital ring homomorphism.

Monday afternoon — material available: Lecture 6 only

Exercise 1. The affine line as a spectrum

★ CORE

Assume that $\mathbb{k}$ is algebraically closed and let $X = \text{Spec}(\mathbb{k}[t])$.

- (i) List the points of X .
- (ii) Determine which points are closed and compute the closure of each point.
- (iii) For a non-zero polynomial f , describe the closed subset $\mathbb{V}(f)$.

Exercise 2. The spectrum of the integers

★ CORE

Describe $\text{Spec}(\mathbb{Z})$: its points, its closed points, and the closure of each point. For $n \neq 0$, determine $\mathbb{V}(n)$.

Exercise 3. Nilpotents are invisible topologically

★ CORE

Let

$$A = \mathbb{k}[\varepsilon]/(\varepsilon^2), \quad X = \text{Spec}(A).$$

Determine the points and open subsets of X . Compare the underlying topological space with $\text{Spec}(\mathbb{k})$, and explain why the two affine schemes are nevertheless different.

Exercise 4. Kernels, images and their closures

★★ CORE

For each homomorphism, compute the kernel and use Proposition II.B.2 to determine the closure of the induced map on spectra.

(i)

$$\varphi : \mathbb{k}[x, y] \longrightarrow \mathbb{k}[t], \quad x \longmapsto t^2, \quad y \longmapsto t^3.$$

(ii)

$$\psi : \mathbb{k}[x, y] \longrightarrow \mathbb{k}[t, t^{-1}], \quad x \longmapsto t, \quad y \longmapsto 0.$$

Determine the image itself and exhibit explicitly the point of its closure which it misses.

(iii)

$$\chi : \mathbb{k}[x, y] \longrightarrow \mathbb{k}[t, t^{-1}], \quad x \longmapsto t, \quad y \longmapsto t^{-1}.$$

Show in this case that the image equals its closure.

Exercise 5. Closed subsets in the node

★★ CHOICE

Let $A = \mathbb{k}[x, y]/(xy)$ and $X = \text{Spec}(A)$. Denote the classes of x, y by the same letters.

- (i) Show that $\mathbb{V}(x) \cup \mathbb{V}(y) = X$ and determine $\mathbb{V}(x) \cap \mathbb{V}(y)$.
- (ii) Compute the closures of the points (x) and (y) .
- (iii) Show that neither of these two points is closed.

Thursday afternoon — material available: Lectures 6–9

Exercise 6. The affine line over several fields

★★ CORE

Let K be a field and let $X = \text{Spec}(K[u])$.

- (i) Describe all points of X , their closures, and their residue fields in terms of irreducible polynomials in $K[u]$.
- (ii) Specialise the description to $K = \mathbb{C}$, $K = \mathbb{R}$, and $K = \mathbb{F}_p$. Which closed points are K -rational?

- (iii) Take $K = \mathbb{C}(t)$. Give a K -rational closed point and a non-rational closed point, and compute their residue fields.

Exercise 7. Principal opens in the node

★★ CORE

Let $A = \mathbb{k}[x, y]/(xy)$ and $X = \text{Spec}(A)$.

- (i) Describe $D(x)$ and $D(y)$ as affine schemes.
- (ii) Show that $D(x) \cap D(y) = \emptyset$ and identify the point missing from $D(x) \cup D(y)$.
- (iii) Compute the residue fields at (x) , (y) and (x, y) .

Exercise 8. Sections and a stalk on the affine line

★ CORE

Let $X = \text{Spec}(\mathbb{k}[t])$ and $U = D(t(t-1))$.

- (i) Compute $\mathcal{O}_X(U)$.
- (ii) Compute the stalk $\mathcal{O}_{X, (t-a)}$ and its residue field.
- (iii) For which a does $(t-a)$ belong to U ?

Exercise 9. A subscheme of the affine line with two support points

★★ CORE

Let

$$Z = \text{Spec}(\mathbb{k}[t]/(t^3(t-1)^2)).$$

- (i) Determine the underlying topological space of Z .
- (ii) Decompose its coordinate ring as a product of two rings.
- (iii) Compare the two local pieces with the reduced points at 0 and 1.

Exercise 10. Prime ideals and scheme structure in the affine plane

★★ CORE

Assume that $\mathbb{k}$ is algebraically closed and work in $A = \mathbb{k}[x, y]$. Put

$$I = (xy), \quad J = (x^2y), \quad Q = (x, y)^2.$$

- (i) Using the description of the prime ideals of A , determine the primes containing I , J , and Q . Hence compare the three underlying closed subsets.
- (ii) Verify in these cases the formula

$$\sqrt{\mathfrak{a}} = \bigcap_{\mathfrak{p} \supseteq \mathfrak{a}} \mathfrak{p}.$$

- (iii) Explain how $\text{Spec}(A/I)$, $\text{Spec}(A/J)$, and $\text{Spec}(A/Q)$ retain different scheme structures even when their points coincide.

Exercise 11. Products of rings and disjoint unions

★★ CHOICE

Let B and C be non-zero rings.

- (i) Show that every prime ideal of $B \times C$ is of the form $\mathfrak{p} \times C$ or $B \times \mathfrak{q}$.
- (ii) Deduce that

$$\text{Spec}(B \times C) \simeq \text{Spec}(B) \amalg \text{Spec}(C).$$

- (iii) Relate this to the idempotents $(1, 0)$ and $(0, 1)$.

After Friday's arithmetic lecture or homework

Exercise 12. Prime ideals in $\mathbb{Z}[\sqrt{3}]$

★★ CHOICE

Let

$$A = \mathbb{Z}[s]/(s^2 - 3), \quad f = \text{Spec}(\gamma_A) : \text{Spec}(A) \longrightarrow \text{Spec}(\mathbb{Z}).$$

For a prime p , describe the points of $\text{Spec}(A)$ mapping to (p) by factoring $s^2 - 3$ over $\mathbb{F}_p$. Treat separately $p = 2$, $p = 3$, and $p \neq 2, 3$.

Exercise 13. An arithmetic conic

★★★ OPTIONAL

Put $A = \mathbb{Z}[x, y]/(x^2 + y^2 - 5)$. For

$$X = \operatorname{Spec}(A) \xrightarrow{\operatorname{Spec}(\gamma_A)} \operatorname{Spec}(\mathbb{Z}),$$

write down the equation obtained after reduction modulo a prime p . Determine for which p that equation is a square or has a singular point.

Algebraic Geometry – Exercise Sheet 3

Week 3: Schemes, morphisms, gluing and first global properties
Monday and Thursday afternoon sessions; Lectures 11–15

Difficulty. ★ routine; ★★ synthesis; ★★★ challenge or extension.

Use. The sheet is deliberately longer than the exercise sessions. Cerulean exercises form the suggested route; green exercises are alternatives.

For a ring R , write $\gamma_R : \mathbb{Z} \rightarrow R$, $n \mapsto n1_R$, for the unique unital ring homomorphism.

Monday afternoon — material available: Lectures 1–11

Exercise 1. Global functions as morphisms to the affine line

★ **CORE**

Let X be a scheme.

- (i) Choose an affine open covering $X = \bigcup_i U_i$. Explain why a global section $s \in \Gamma(\mathcal{O}_X)$ determines compatible morphisms $U_i \rightarrow \mathbb{A}_{\mathbb{Z}}^1$ and hence a morphism

$$f_s : X \longrightarrow \mathbb{A}_{\mathbb{Z}}^1.$$

- (ii) Show that

$$\mathrm{Hom}(X, \mathbb{A}_{\mathbb{Z}}^1) \simeq \Gamma(\mathcal{O}_X).$$

- (iii) If $X = \mathrm{Spec}(A)$, describe the morphism associated with $a \in A$.

Exercise 2. The reduced point and the double point

★ **CORE**

Inside $\mathbb{A}_{\mathbb{k}}^1 = \mathrm{Spec}(\mathbb{k}[t])$, consider

$$P = \mathrm{Spec}(\mathbb{k}[t]/(t)), \quad D = \mathrm{Spec}(\mathbb{k}[t]/(t^2)).$$

- (i) Show that the two closed immersions have the same image as subsets of $\mathbb{A}_{\mathbb{k}}^1$.
(ii) Show that P and D are not isomorphic as schemes.
(iii) Compute the maps on local rings at their unique points.

Exercise 3. A morphism between affine schemes

★ **CORE**

Let

$$\varphi : \mathbb{k}[u, v] \longrightarrow \mathbb{k}[t], \quad u \longmapsto t, \quad v \longmapsto t^2.$$

- (i) Describe the induced morphism $f : \mathbb{A}_{\mathbb{k}}^1 \rightarrow \mathbb{A}_{\mathbb{k}}^2$.
(ii) Determine $\ker(\varphi)$ and factor f through a closed subscheme of $\mathbb{A}_{\mathbb{k}}^2$.
(iii) Show that the resulting morphism onto that closed subscheme is an isomorphism.

Exercise 4. Gluing ideal sheaves and closed subschemes

★★ **CORE**

Let $X = \bigcup_i U_i$ be an open covering. Suppose that each U_i carries a quasi-coherent ideal sheaf $\mathcal{I}_i \subset \mathcal{O}_{U_i}$ and that

$$\mathcal{I}_i|_{U_i \cap U_j} = \mathcal{I}_j|_{U_i \cap U_j}$$

inside $\mathcal{O}_X|_{U_i \cap U_j}$.

- (i) Show that the $\mathcal{I}_i$ glue to an ideal sheaf $\mathcal{I} \subset \mathcal{O}_X$.
(ii) Prove that $\mathcal{I}$ is quasi-coherent.
(iii) If $Z_i \hookrightarrow U_i$ is the closed subscheme defined by $\mathcal{I}_i$, show that the Z_i glue to the closed subscheme of X defined by $\mathcal{I}$.
(iv) Explain why none of these assertions requires X to be affine.

Exercise 5. Open subschemes and the punctured plane

★★ **CORE**

Let X be a scheme and let $U \subset X$ be open.

- (i) Prove directly from the locally affine definition that $(U, \mathcal{O}_X|_U)$ is a scheme.

(ii) Show that for every $x \in U$ one has

$$\mathcal{O}_{U,x} \simeq \mathcal{O}_{X,x}, \quad \kappa_U(x) \simeq \kappa_X(x).$$

(iii) Cover the punctured affine plane $P = \mathbb{A}_{\mathbb{k}}^2 \setminus \mathbb{W}(x, y)$ by two affine open subschemes.

(iv) Compute $\Gamma(\mathcal{O}_P)$ and prove that P is not affine.

Thursday afternoon — material available: Lectures 1–14

Exercise 6. Dual numbers and tangent vectors

★★ CORE

Let $D = \operatorname{Spec}(\mathbb{k}[\varepsilon]/(\varepsilon^2))$ and let $p = (a, b) \in \mathbb{A}_{\mathbb{k}}^2(\mathbb{k})$.

(i) Show that morphisms $D \rightarrow \mathbb{A}_{\mathbb{k}}^2$ whose closed point maps to p are uniquely of the form

$$x \mapsto a + u\varepsilon, \quad y \mapsto b + v\varepsilon.$$

(ii) Let $C = \mathbb{W}(f) \subset \mathbb{A}_{\mathbb{k}}^2$. Show that such a morphism factors through C precisely when

$$f(a, b) = 0, \quad f_x(a, b)u + f_y(a, b)v = 0.$$

(iii) Apply this to the parabola $y = x^2$.

(iv) Assume $\operatorname{char}(\mathbb{k}) \neq 2, 3$. For the cusp $C = \mathbb{W}(y^2 - x^3)$, compare the tangent spaces at $(0, 0)$ and $(1, 1)$. Which point is singular and which is smooth according to the Jacobian criterion?

Exercise 7. The affine plane with a doubled line

★★ CORE

Let

$$U_1 = \operatorname{Spec}(\mathbb{k}[x_1, y_1]), \quad U_2 = \operatorname{Spec}(\mathbb{k}[x_2, y_2]).$$

Glue $D(x_1) \subset U_1$ to $D(x_2) \subset U_2$ by $x_1 \mapsto x_2$, $y_1 \mapsto y_2$, and call the resulting scheme X .

(i) Describe the underlying topological space: which points occur twice?

(ii) Show that $\Gamma(\mathcal{O}_X) \simeq \mathbb{k}[x, y]$.

(iii) Show that the affinisation map $X \rightarrow \operatorname{Spec} \Gamma(\mathcal{O}_X)$ identifies the two copies of the line $x = 0$. Deduce that X is not affine.

(iv) Show that X is connected.

(v) After separatedness has been introduced, use the affine-cover criterion to prove that X is not separated over $\mathbb{k}$.

Exercise 8. Rational and closed points of the projective line

★★ CORE

Construct $\mathbb{P}_{\mathbb{k}}^1$ by gluing $U_0 = \operatorname{Spec}(\mathbb{k}[t])$ and $U_1 = \operatorname{Spec}(\mathbb{k}[s])$ along $s = t^{-1}$.

(i) For every field extension $K/\mathbb{k}$, prove that

$$\mathbb{P}_{\mathbb{k}}^1(K) = \mathbf{P}_K^1.$$

Describe the points in the two standard charts.

(ii) Describe all closed points of $\mathbb{P}_{\mathbb{k}}^1$, compute their residue fields, and determine which are $\mathbb{k}$ -rational.

(iii) For every closed point x , exhibit its canonical $\kappa(x)$ -valued point and verify that $\kappa(x)/\mathbb{k}$ is finite. Give a field-valued point whose image is not closed.

(iv) Prove that $\Gamma(\mathcal{O}_{\mathbb{P}_{\mathbb{k}}^1}) = \mathbb{k}$.

(v) Deduce that $\mathbb{P}_{\mathbb{k}}^1$ is not affine.

Exercise 9. The projective conic from its standard opens

★★ CORE

Let

$$S = \mathbb{k}[x_0, x_1, x_2]/(x_0x_2 - x_1^2), \quad C = \operatorname{Proj}(S).$$

(i) Compute $(S[x_0^{-1}])_0$ and $(S[x_2^{-1}])_0$.

- (ii) Identify $D_+(x_0)$ and $D_+(x_2)$ with affine lines.
- (iii) Describe their overlap and recover the gluing rule for $\mathbb{P}^1$.
- (iv) Show that projection from $p = [1 : 0 : 0]$ extends to a morphism $\pi : C \rightarrow \mathbb{P}_{\mathbb{k}}^1$ by using the formulas $[x_0 : x_1]$ on $D_+(x_0)$ and $[x_1 : x_2]$ on $D_+(x_2)$. Prove that its inverse is $[s : t] \mapsto [s^2 : st : t^2]$.

Exercise 10. Base loci: projection and Cremona

*** CHOICE

- (i) Consider

$$\pi : \mathbb{P}_{\mathbb{k}}^2 \dashrightarrow \mathbb{P}_{\mathbb{k}}^1, \quad [x_0 : x_1 : x_2] \longmapsto [x_1 : x_2].$$

Determine its domain of definition and its base locus.

- (ii) In $\mathbb{P}_{\mathbb{k}}^2 \times \mathbb{P}_{\mathbb{k}}^1$, show that the closure of its graph is defined by $x_1v - x_2u = 0$. Describe the fibre of the first projection over $[1 : 0 : 0]$.
- (iii) For the standard Cremona transformation

$$[x_0 : x_1 : x_2] \dashrightarrow [x_1x_2 : x_0x_2 : x_0x_1],$$

compute the base locus and show that the transformation is its own inverse on the dense torus $x_0x_1x_2 \neq 0$.

Exercise 11. Connectedness beyond affine schemes

** CORE

Let X be a scheme.

- (i) Prove that X is disconnected if and only if $\Gamma(\mathcal{O}_X)$ has a non-trivial idempotent.
- (ii) Use the global-function computations from this sheet to show that $\mathbb{P}_{\mathbb{k}}^1$ and the affine plane with a doubled line are connected.
- (iii) Compare the connected reducible scheme $\text{Spec}(\mathbb{k}[x, y]/(xy))$ with the disconnected scheme $\text{Spec}(\mathbb{k} \times \mathbb{k})$.

Exercise 12. A doubled projective line and its reduction

*** CORE

Let

$$U_0 = \text{Spec}(\mathbb{k}[t, \varepsilon]/(\varepsilon^2)), \quad U_1 = \text{Spec}(\mathbb{k}[s, \eta]/(\eta^2)).$$

Glue $D(t) \subset U_0$ to $D(s) \subset U_1$ using $s = t^{-1}$ and $\eta = \varepsilon$, and call the resulting scheme X .

- (i) Show that $(\varepsilon) \subset \mathcal{O}_{U_0}$ and $(\eta) \subset \mathcal{O}_{U_1}$ glue to a quasi-coherent ideal sheaf $\mathcal{I} \subset \mathcal{O}_X$ with $\mathcal{I}^2 = 0$.
- (ii) Describe X_{red} and prove that $X_{\text{red}} \simeq \mathbb{P}_{\mathbb{k}}^1$.
- (iii) Show that every morphism from a reduced scheme Y to X factors uniquely through $X_{\text{red}} \hookrightarrow X$.
- (iv) Deduce that X is not affine.

Exercise 13. A weighted projective chart

*** OPTIONAL

Let $R = \mathbb{k}[x, y, z]$ with $\deg x = \deg y = 1$ and $\deg z = 2$, and set $X = \text{Proj}(R)$. Show that $D_+(x)$ and $D_+(y)$ are affine planes and that

$$D_+(z) \simeq \text{Spec}(\mathbb{k}[a, b, c]/(ac - b^2)).$$

Identify the singular point of this chart.

After Friday's lecture or preparation for Week 4

Exercise 14. Components and generic points

** CHOICE

Let

$$I = (xy, xz) \subset \mathbb{k}[x, y, z], \quad X = \text{Spec}(\mathbb{k}[x, y, z]/I).$$

Show that $I = (x) \cap (y, z)$. Determine the irreducible components, their generic points and dimensions, and decide whether X is connected or integral.

Exercise 15. The function field of the cusp

** CHOICE

Let

$$C = \text{Spec}(\mathbb{k}[x, y]/(y^2 - x^3)).$$

Show that C is integral and that $K(C) \simeq \mathbb{k}(t)$.

Algebraic Geometry – Exercise Sheet 4

Week 4: Generic points, dimension and scheme-theoretic fibres
One Monday exercise session, followed by a revision bank; Lectures 15–18

Difficulty. ★ routine; ★★ synthesis; ★★★ challenge or extension.

Use. The sheet is deliberately longer than the exercise sessions. Cerulean exercises form the suggested route; green exercises are alternatives.

Monday afternoon — material available: Lectures 1–16

Exercise 1. Components and generic points

★★ CORE

Let

$$I = (xy, xz) \subset \mathbb{k}[x, y, z], \quad X = \operatorname{Spec}(\mathbb{k}[x, y, z]/I).$$

- (i) Show that $I = (x) \cap (y, z)$.
- (ii) Determine the irreducible components and their generic points.
- (iii) Is X reduced, connected, irreducible, or integral?

Exercise 2. The function field of the cusp

★★ CORE

Let

$$C = \operatorname{Spec}(\mathbb{k}[x, y]/(y^2 - x^3)).$$

Show that C is integral, compute its function field, and deduce its dimension.

Exercise 3. Chains and dimensions

★ CORE

- (i) Exhibit a maximal chain of irreducible closed subsets in $\operatorname{Spec}(\mathbb{Z})$.
- (ii) Give a chain of length n in $\mathbb{A}_{\mathbb{k}}^n$.
- (iii) Determine the dimension of

$$\operatorname{Spec}(\mathbb{k}[x, y, z]/(xy, xz))$$

from its components.

Exercise 4. Noether normalisation and dimension

★★ CORE

Let

$$B = \mathbb{k}[x, y]/(y^2 - x^3).$$

- (i) Show that B is finite over $\mathbb{k}[x]$.
- (ii) Use this to recover $\dim(B) = 1$.
- (iii) Compare with the computation from the function field.

After Lectures 17–18 — revision bank on fibres

Exercise 5. The generic affine line

★ CORE

Let

$$\pi : \mathbb{A}_{\mathbb{C}}^2 = \operatorname{Spec}(\mathbb{C}[t, x]) \longrightarrow \mathbb{A}_{\mathbb{C}}^1 = \operatorname{Spec}(\mathbb{C}[t])$$

be the projection. Compute its fibre over a closed point $(t - a)$ and its fibre over the generic point (0) . Explain why the latter is $\operatorname{Spec}(\mathbb{C}(t)[x])$ and verify the dimension formula for the generic fibre.

Exercise 6. A fibre product larger than the set-theoretic one

★★ CHOICE

Let $X = \operatorname{Spec}(\mathbb{C})$ and $S = \operatorname{Spec}(\mathbb{R})$. Compute $X \times_S X$ and compare its points with the set-theoretic fibre product of the underlying one-point spaces.

Exercise 7. The power map and its fibres

★★ CORE

Assume that $\mathbb{k}$ is algebraically closed of characteristic 0. Let

$$f : \mathbb{A}_{\mathbb{k}}^1 \longrightarrow \mathbb{A}_{\mathbb{k}}^1, \quad t \longmapsto t^n.$$

Compute every scheme-theoretic fibre, determine when it is reduced, compute the generic fibre, and show that f is finite.

Exercise 8. A family whose special fibre becomes reducible

★★ **CORE**

Let

$$X = \operatorname{Spec}(\mathbb{k}[x, y, t]/(xy - t)) \longrightarrow \mathbb{A}_t^1.$$

Compute the fibres, show that all have dimension 1, and determine which are integral and which are reducible.

Exercise 9. A jumping fibre

★★ **CORE**

Consider

$$f : \mathbb{A}_{\mathbb{k}}^2 \longrightarrow \mathbb{A}_{\mathbb{k}}^2, \quad (x, y) \longmapsto (x, xy).$$

Compute the fibre over (a, b) and show that the generic fibre has dimension 0, while the fibre over $(0, 0)$ has dimension 1.

Exercise 10. The generic fibre formula in examples

★★ **CHOICE**

Verify

$$\dim X_\eta = \dim X - \dim Y$$

for the map in the preceding exercise and for the projection $\mathbb{A}_{\mathbb{k}}^{m+n} \rightarrow \mathbb{A}_{\mathbb{k}}^m$.

Alternative route: separatedness and properness

Exercise 11. The valuative criterion: affine versus projective space

★★ **CORE**

Let $R = \mathbb{k}[[\pi]]$ and $K = \mathbb{k}((\pi))$.

- (i) Show that R is a valuation ring and describe $\operatorname{Spec}(R)$.
- (ii) Use the K -point $\pi^{-1} \in \mathbb{A}_{\mathbb{k}}^1(K)$ to prove that $\mathbb{A}_{\mathbb{k}}^1$ is not proper over $\mathbb{k}$.
- (iii) Given $[a_0 : \cdots : a_n] \in \mathbb{P}_{\mathbb{k}}^n(K)$, multiply the coordinates by a common power of π so that all lie in R and at least one is a unit. Deduce that the point extends to an R -point of $\mathbb{P}_{\mathbb{k}}^n$.
- (iv) Explain where separatedness supplies uniqueness and conclude, using the valuative criterion, that $\mathbb{P}_{\mathbb{k}}^n$ is proper.

Exercise 12. Global functions on a proper integral variety

★★★ **CORE**

Let X be an integral proper scheme of finite type over a field $\mathbb{k}$, and put $B = \Gamma(\mathcal{O}_X)$. For $b \in B$:

- (i) construct the corresponding morphism $g_b : X \rightarrow \mathbb{A}_{\mathbb{k}}^1$;
- (ii) put $C = \operatorname{im}(\mathbb{k}[t] \rightarrow B)$ and $Z = \operatorname{Spec}(C)$. Show that g_b factors as a dominant morphism $X \rightarrow Z$ followed by a closed immersion $Z \hookrightarrow \mathbb{A}_{\mathbb{k}}^1$;
- (iii) use properness to show that Z is proper over $\mathbb{k}$, then use the fact that an affine proper morphism is finite to prove that b is algebraic over $\mathbb{k}$;
- (iv) let $\mathbb{k}_X$ be the algebraic closure of $\mathbb{k}$ inside the finitely generated field $K(X)$. Prove that

$$B \subseteq \mathbb{k}_X,$$

and conclude that B is finite-dimensional over $\mathbb{k}$. What happens when $\mathbb{k}$ is algebraically closed?

Optional outreach exercises: Hilbert schemes of points

Exercise 13. $\operatorname{Hilb}^n(\mathbb{A}^1)$

★★ **CHOICE**

Assume that $\mathbb{k}$ is algebraically closed. Show that every ideal $I \subset \mathbb{k}[x]$ with $\dim_{\mathbb{k}} \mathbb{k}[x]/I = n$ is generated by a unique monic polynomial of degree n . Deduce

$$\operatorname{Hilb}^n(\mathbb{A}_{\mathbb{k}}^1) \simeq \mathbb{A}_{\mathbb{k}}^n$$

and explain how factorisation records support and multiplicities.

Exercise 14. Commuting matrices and a cyclic vector

★★★ **OPTIONAL**

Let V be n -dimensional and let (X, Y, v) satisfy $[X, Y] = 0$ and $\mathbb{k}[X, Y]v = V$. Show that

$$I = \{f \in \mathbb{k}[x, y] \mid f(X, Y)v = 0\}$$

is an ideal and $\mathbb{k}[x, y]/I \simeq V$. Determine the triple for $I = (y, x^2)$.